

महाराष्ट्र स्थापत्य अभियांत्रिकी सेवा मुख्य परीक्षा - 2024
दिनांक - 9 मे, 2024.



2024

E20

BOOKLET NO.

600401

**Civil Engineering
Paper – I**

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Questions no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Sections.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note – 1. Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

SEAL



SECTION - A

Q1. Answer **any five** of the following :

(8×5=40)

- (a) A 12 mm diameter steel rod is welded to a rigid plate that is supported by a brass pipe whose outside diameter is 30 mm and inside diameter is 20 mm as shown in Figure No. 1.1. If $P = 10 \text{ kN}$, $E_s = 210 \text{ GPa}$ and $E_b = 105 \text{ GPa}$. Calculate :
- The displacement of point A.
 - The stress in steel and brass.

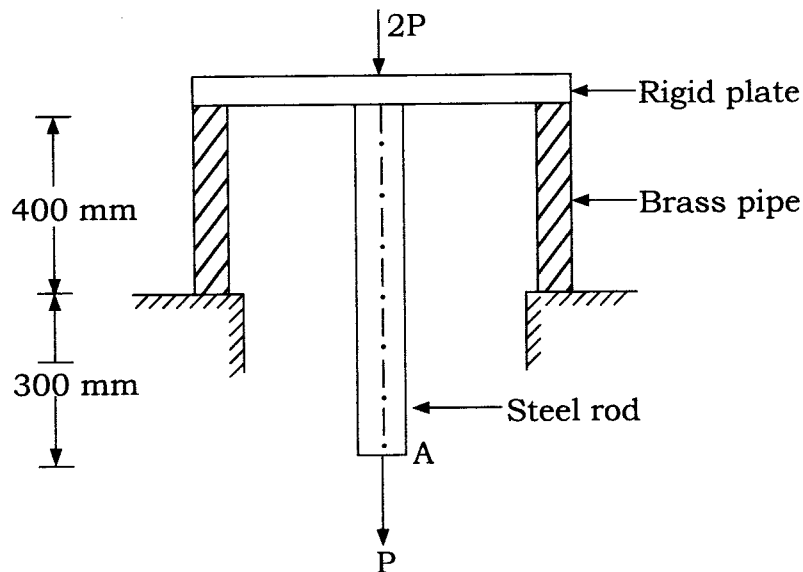


Figure 1.1

- (b) A continuous beam ABC is simply supported at A and C. Span AC and BC are of lengths l_1 and l_2 respectively. The beam is loaded with W_1 and W_2 acting at centres of AB and BC respectively. Calculate support moment M_B and support reactions R_A , R_B and R_C if $l_1 = l_2 = l$ and $W_1 = W_2 = W$.

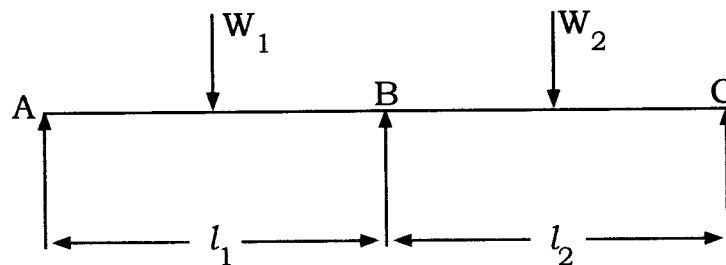


Figure 1.2



- (c) Develop stiffness matrix for a prismatic beam member with both ends fixed.
- (d) Explain the effect of temperature change in a two hinged arch.
- (e) A propped cantilever ABCD is loaded as shown in Figure 1.3. Calculate the collapse load if the beam is of uniform cross section.

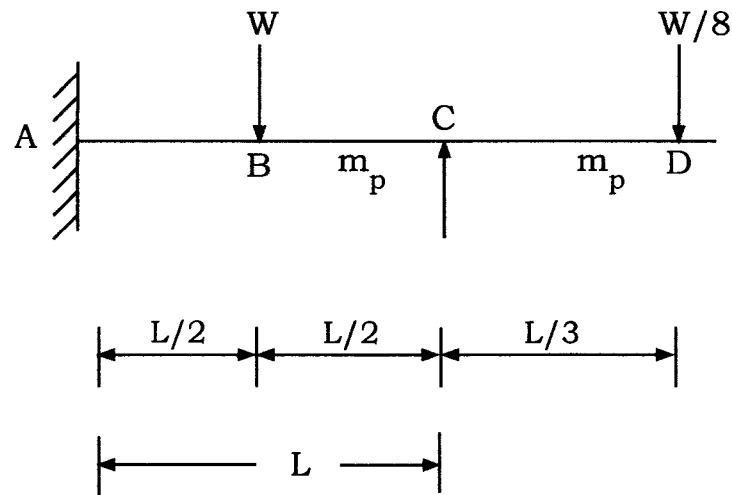


Figure No. 1.3

- (f) A grader will be used to maintain 2.5 miles of haul road. It is estimated this work will required three passes in first gear, levelling the road requires two passes in second gear and final shaping of the road requires two passes in fourth gear. Use an efficiency factor of 60%. Grader speeds are given in the table. Estimate the time requirement for this work.

Sr.No.	Forward gear	Maximum travel speed (mph)	Remark
01	First	2.3	
02	Second	3.2	
03	Third	4.4	
04	Fourth	6.4	

- (g) Explain Euler's Theory of Buckling of Columns. Derive equation for critical or Euler load for column hinged at both ends and subjected to axial compression load P.



- Q2.** (a) A hollow circular steel pole is vertical and 6.0 m high and fixed at the bottom as shown in Figure 2.1. It is made in 2 sections. The top 1.5 m having a moment of inertia equal to 4680 cm^4 and lower 4.50 m having a moment of inertia of 6600 cm^4 . The horizontal pull of the pole is counter acted by a force of 25 kN in the stay located as shown in Figure. Calculate the deflection of the pole at top. Neglect all other deformations except those due to bending action. E for steel $E_s = 210 \text{ GPa}$. **15**

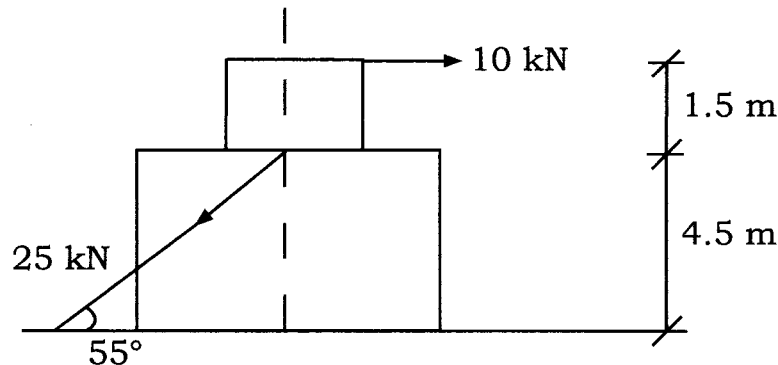


Figure 2.1

- (b) Analyse the frame shown in figure 2.2 by moment distribution method. Take M.I. for BC and AD – $2I$ and AB and DC – I . **15**

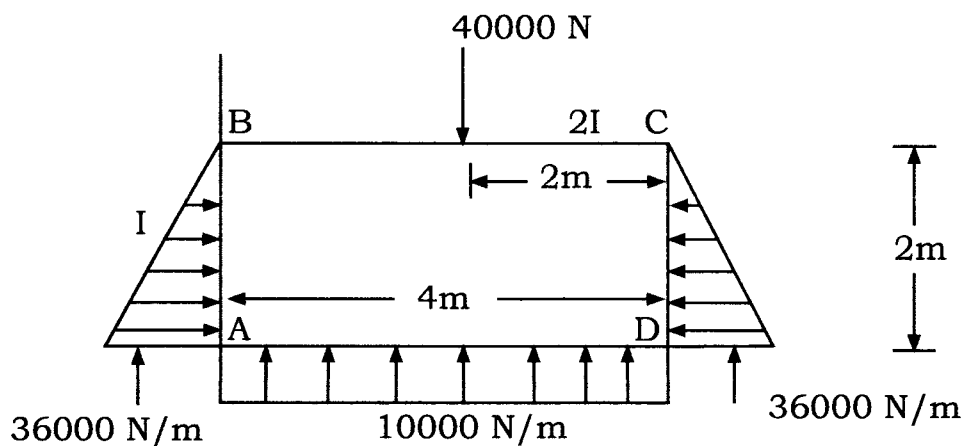


Figure 2.2

- (c) List different iteration methods for solving linear equations. Explain in detail one method. Solve following equations by any suitable method. **10**
- $$\begin{aligned} 20x + y - 2z &= 17 \\ 3x + 20y - z &= -18 \\ 2x - 3y + 20z &= 25 \end{aligned}$$
-



- Q3.** (a) The two span continuous beam shown in Figure 3.1 is subjected to a uniform load W (equal to $4P/L$) in span AB and two concentrated forces P in span BC. Both members have the same length L , but the flexural rigidity of member AB is twice that of member BC. **15**
Calculate end-actions, support reactions and joint displacements for the beam.

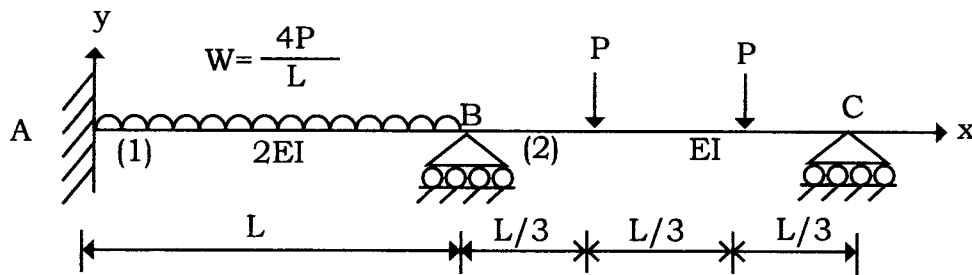


Figure 3.1

- (b) A deck type plate girder railway bridge is to be constructed for a broad gauge single line track on the main line data is given below. **15**
- i) Effective span = 20 m
 - ii) c/x distance between plate girder = 2 m
 - iii) Dead load on each girder = $220 L + 600$ N/m
 - iv) Dead load of track with sleepers = 6800 N/m
 - v) Lateral load = 9000 N/m
 - vi) For 20 m span and B.G.
- Total load for bending moment – 1964 kN
Total load for shear – 2168 kN
CDA $(0.15 + 8/6 + L) - 0.458$
Design the superstructure of the bridge.
- (c) Explain various ratios such as materials cost indexes, vendor rating indexes, material planning and forecasting indexes, efficiency indexes, inventory and stores indexes. Transportation and distribution indexes and standardisation, condensation and value analysis. Which can be used to evaluate the performance of different functions of material management. **10**



- Q4.** (a) Explain individual and group incentive schemes. Compare different incentive plans.

The following information is given for an operator's output on a machine for six days.

Day	Mon.	Tues.	Wed.	Thu.	Fri.	Sat.
Output	50	55	48	40	35	45

The incentive scheme is as follows.

Output below standard – Time wages

Output at standard – Time wages + 20% bonus

Output above standard – Higher piece rate on entire production

The hourly rate of the operator is Rs. 100. The standard time is 12 min. per piece of the job and higher rate is Rs. 25 per piece. Calculate daywise earnings of the operator.

15

- (b) Analyse the continuous beam shown in Figure 4.1 by the slope deflection method. The supports are at same level and the beam is of constant stiffness throughout.

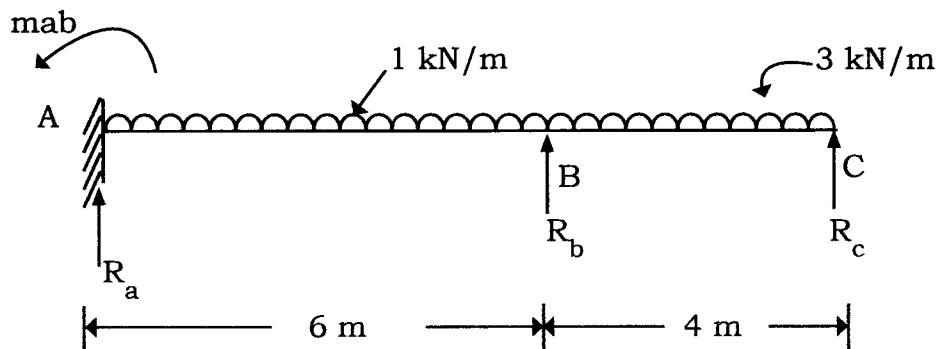
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Figure 4.1

- (c) Determine the influence lines for the shear force and bending moment at D, the middle point of span BC of a continuous beam shown in Figure 4.2. Compute ordinates at 2 m interval.

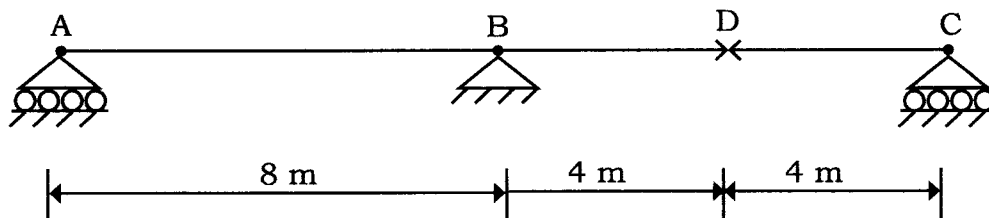
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Figure 4.2

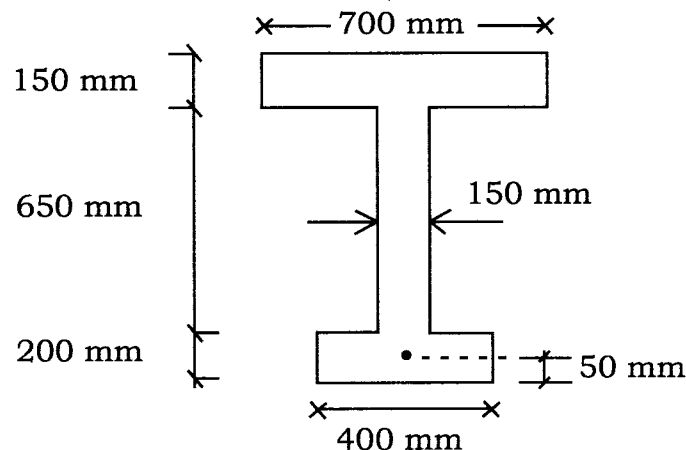
**SECTION - B**

- Q5.** Answer **any five** of the following. **(8×5=40)**
- (a) Use limit state method to design a reinforced concrete beam for flexure only to support a uniformly distributed services load of 25 kN/m, over an effective S.S. span of 6m. The materials used are M 20 grade concrete and HYSD steel of grade Fe415 ; Assume $d = 2b$.
 - (b) Explain in details types, parts and design procedure of retaining wall.
 - (c) Explain the classification of bridges based on material, span and purpose.
 - (d) Explain in details the test carried out on wet and hardened concrete.
 - (e) Explain in details losses in prestressed concrete as per I.S. code.
 - (f) A sample of saturated soil has a water content of 25% and bulk unit wt. of 20 kN/m³. Determine dry density, void ratio and specific gravity of solid particles. What would be the bulk unit wt. of the same soil at the same void ratio but at a degree of saturation of 80% ? Take $\gamma_w = 10 \text{ kN/m}^3$.
 - (g) Write a note on "Scaffolding" and draw a sketch of formwork for slab and beam system.
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- Q6.** (a) Design two way slab of clear size 3 m × 4.5 m, live load = 2.5 kN/m². Slab is simply supported on four sides with corner free to lift. Side support beam width is 225 mm. Use concrete M15 and Fe415 steel. **15**
- (b) Design dog legged stair for building in which the vertical distance between the floor is 3.6 m. The stair hall measures 2.5 m × 5.0 m. The live load may be taken as 2500 N/m². Use M20 concrete and Fe415 reinforcement. **15**
- (c) Design an isolated column footing of uniform thickness under load 700 kN acting on column size 450 mm × 450 mm. SBC of soil is 200 kN/m². Materials used M20 and Fe415. **10**
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- Q7.** (a) Write short notes on : **15**
- 1) Afflux in bridge engineering.
 - 2) Loads and forces on bridges.
 - 3) Methods of strengthening Cofferdams.
- (b) Explain in details step by step procedure of mix design with example. **15**
- (c) A 5 m high retaining wall with its back surface smooth and vertical retains a soil with its surface horizontal having a cohesion of 5 kN/m² and angle of internal friction of 30°. Determine the Rankin's active earth pressure and its point of application on the wall.
- a) Before the formation of crack b) After the formation of crack.
- Consider unit wt. of soil as 17.5 kN/m³. **10**
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P.T.O.



- Q8.** (a) An unsymmetrical I-section beam is shown in figure. Span of beam is 20 m and it carries live load of 30 kN/m. Beam is prestressed 12/7 fryssinet cables with initial stress of 1200 N/mm^2 . Cable is located at 50 mm from the bottom. Calculate stresses at mid span section of beam at top and bottom under following conditions.
- i) Prestress + dead load ii) Prestress + dead load + live load.
- Assume 15% loss in prestress. **15**



- (b) Explain the concept of dry and wet of optimum moisture content and discuss its effect on properties of soil.

Determine the minimum size of square footing for carrying safe load of 1250 kN. The footing is placed 1.1 m below ground level in a homogeneous sandy soil having unit wt. as 18 kN/m^3 and angle of frictional resistance as 30° . Take factor of safety as 2.5.

For $\phi = 30^\circ$, $N_c = 37.2$, $N_q = 22.5$, $N_\gamma = 19.7$. If the size of footing is restricted to $1.7 \text{ m} \times 1.7 \text{ m}$. Determine the depth of footing. **15**

- (c) Classify different methods of prestressing. Explain in details concept of partial prestressed concrete. **10**